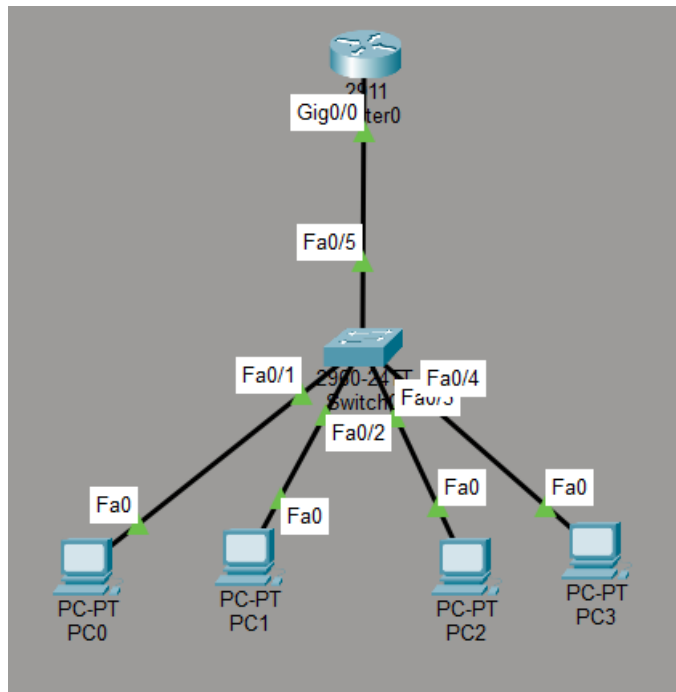


Task 2: Configure an 802.1Q Trunk Link Establish a trunk connection so traffic for multiple VLANs can pass between the switch and router. (Applied — one concept combined with port-mode configuration.) Configure the switch port connecting to the router as an 802.1Q trunk. Restrict the trunk to allow only VLAN 10 and VLAN 20. Verify trunk status using show interfaces trunk. Confirm there is no native VLAN mismatch between the switch and router.



Router

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#

```

Switch

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name team
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name hr
Switch(config-vlan)#exit
Switch(config)#int f0/1-2
      ^
% Invalid input detected at '^' marker.

Switch(config)#int range f0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range f0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#int f0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Fa0/5     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/5     10,20

Port      Vlans allowed and active in management domain
Fa0/5     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/5     10,20

Switch#
```